La Scuola d'Italia / Dr. Huson / IB Math 12 HL Name:
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12.2 Classwork: Complex Numbers Restart (no calculator)

Answer both problems in the spaces provided. Show your working. Give all answers as exact values (surds and fractions of π), not decimals.

1. Let $z = 3 - 2i$ and $w = 1 + 4i$.

   (a) Find zw, giving your answer in the form $a + bi$, where $a, b \in \mathbb{R}$.

   (b) Find $\dfrac{z}{w}$, giving your answer in the form $a + bi$, where $a, b \in \mathbb{R}$.

   (c) Find the exact value of $|z|$.

   (d) Find the real numbers x and y such that $(x + yi)(2 - i) = 7 + 4i$.

2. Consider the equation $z^2 - 4z + 13 = 0$, where $z \in \mathbb{C}$.

   (a) Solve the equation, giving both roots in the form $a + bi$, where $a, b \in \mathbb{R}$.

   (b) Explain why the two roots are complex conjugates of each other.

   The complex number $2 + i\sqrt{3}$ is one root of a *different* quadratic equation with real coefficients.

   (c) Without solving any equation, find the sum and the product of the two roots of that equation.

   (d) Hence write down a quadratic equation with real coefficients that has $2 + i\sqrt{3}$ as a root.